Senior Embedded Linux & Kernel Engineer

Thomas Perrot

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Blog | Github | Gitlab | GnuPG | Linkedin

Embedded Linux and low-level software engineer with 15 years of experience bringing up platforms and delivering production systems. Expertise across Linux kernel and BSP development, boot chains, Yocto/OpenEmbedded, secure systems, and over-the-air updates for ARM, PowerPC, and RISC-V platforms. Open-source contributor to Linux kernel, U-Boot, OpenEmbedded/Yocto, and Buildroot projects.

Technical Expertise

Programming Languages: C, Assembly, Python, Bash, Rust, VHDL

Linux Kernel & Systems: Linux Kernel, U-Boot, Device Drivers, BSP Development, Board Bring-Up, Kernel Debugging

Build Systems: Yocto Project, OpenEmbedded, Buildroot, kas, repo, bitbake

Boot Security & Trust: OP-TEE, TPM 2.0, TF-A, Trusted Boot, Measured Boot, Secure Boot, Encryption, Integrity Verification, AT91

Architectures & Platforms: ARMv7, ARMv8, RISC-V, x86-64, PowerPC, QorIQ, ARMADA, i.MX, STM32MP1, SAMA5, SAMA7, Tegra, U740, X280

Debugging & Testing: QEMU, GDB, Valgrind, Lauterbach TRACE32, Saleae Logic Analyzer, Podman

Standards & Compliance: DO-178B DAL A, EABI, ISO C99, POSIX

Languages

French: Native

English: Professional Working Proficiency

Education

September 2006 – July 2011: Master's Degree (BAC+5, with honors), Architecture of Critical and Real-Time Embedded Systems

Additional Information

Linux technologist and free-software enthusiast. Personal interests include robotics, IoT, retro computing, arcade systems, CrossFit and Hyrox.

Experiences

December 2020 – Present: Embedded Linux & Kernel Engineer

Bootlin, Toulouse, France (EU) · Hybrid

- Port Linux to new platforms and develop device drivers.
- Integrate open-source components and build complete embedded Linux systems.
- Contribute board support, device drivers, and build-system changes upstream to Linux, U-Boot, OpenEmbedded/Yocto, and Buildroot projects.

July 2015 – December 2020: Embedded Linux Engineer

SIGFOX, Toulouse, France (EU) · Hybrid

- Performed board bring-up and developed BSPs, including bootloaders and Linux kernels.
- Delivered OTA update infrastructure with atomic rollback mechanisms for field-deployed devices.
- Developed security stack including TPM 2.0 integration, encryption, integrity verification, trusted boot, and measured boot compliance.

April 2014 – July 2015: Android System Integration Engineer

Intel Android SI NPI, via Celad & SHT, Toulouse, France (EU) · On-site

- Introduced and maintained pre-silicon platforms in Android mainline.
- Published releases, documentation and guidelines.
- Provided bring-up, build, first-boot, and power-on support for engineering teams.

February 2014 – March 2014: ROS System Integration Engineer

Sogeti High Tech, R&D, Toulouse, France (EU) · On-site

- Designed and implemented an embedded ROS node with ROSc for coOS.
- Designed and implemented a custom ROS bridge for Unicom framework.

August 2011 – February 2014: Embedded System Software Engineer

Airbus S.A.S, via Sogeti High Tech, Toulouse, France (EU) · On-site

- Developed and maintained primary flight-control software under DO-178B certification.
- Optimized embedded code WCET and performed static certification analysis.

February 2011 – August 2011: Research Assistant / ROS System Developer

Five-month contract at **Office National d'Études et de Recherches Aérospatiales**, Toulouse, France (EU) · On-site

- Designed a ROS tool to simulate communication between different software modules of heterogeneous critical systems.

October 2010 – February 2011: Internship / VHDL Processor Designer

Five-month internship at **Atelier Interuniversitaire de Micro-nano Électronique**, Toulouse, France (EU) · On-site

- Designed a 32-bit scalar, pipelined, in-order RISC processor with a Harvard architecture.

February 2010 – August 2010: Research Assistant – WCET Analysis

Six-month contract at **Institut de Recherche en Informatique de Toulouse**, Toulouse, France (EU) · On-site

- Performed timing analysis of parallel processes on multi-core processors.